Health Deterioration Detection in Hypertensives

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Introduction

Sentinel Healthcare provides a remote management platform for clinicians and hospitals to monitor patient health and provide outreach.

Patients regularly record vital signs with monitoring devices for analysis by Sentinel.

UW capstone team developed prediction models for based on Sentinel's vital signs, medication, and clinical notes data to improve health deterioration detection [1].

Data Analysis

Three anonymized datasets provided by Sentinel:

340,967 patient vitals for 1649 patients, containing Systolic Blood Pressure, Diastolic Blood Pressure, and Heart Rate readings from 11/18/2018 to 01/14/2020. After cleaning, 883 remaining patients.

Clinical notes written during each contact with a patient enrolled in the program. Notes contain information ranging from doctors' updates to documentation of health deterioration or hospitalization.15 patients had notes indicating a critical event.

Patient prescription of Beta Blockers, a class of medication for Hypertensives that reduces HR and BP (not applied for this analysis).

Data Processing

The following data pre-processing rules used:

1. Each patient modeled independently with data aggregated to average.

2. Each data series must contain >= 30 days continuous data; only most recent data used.

3. Data series with no variation (σ=0) not considered (due to likely sample error).

4. Missing segments <= 15 days are interpolated.

Models

Three models to predict SBP and HR:

1. Univariate LSTM model built with Keras and TensorFlow with a SciPy environment.

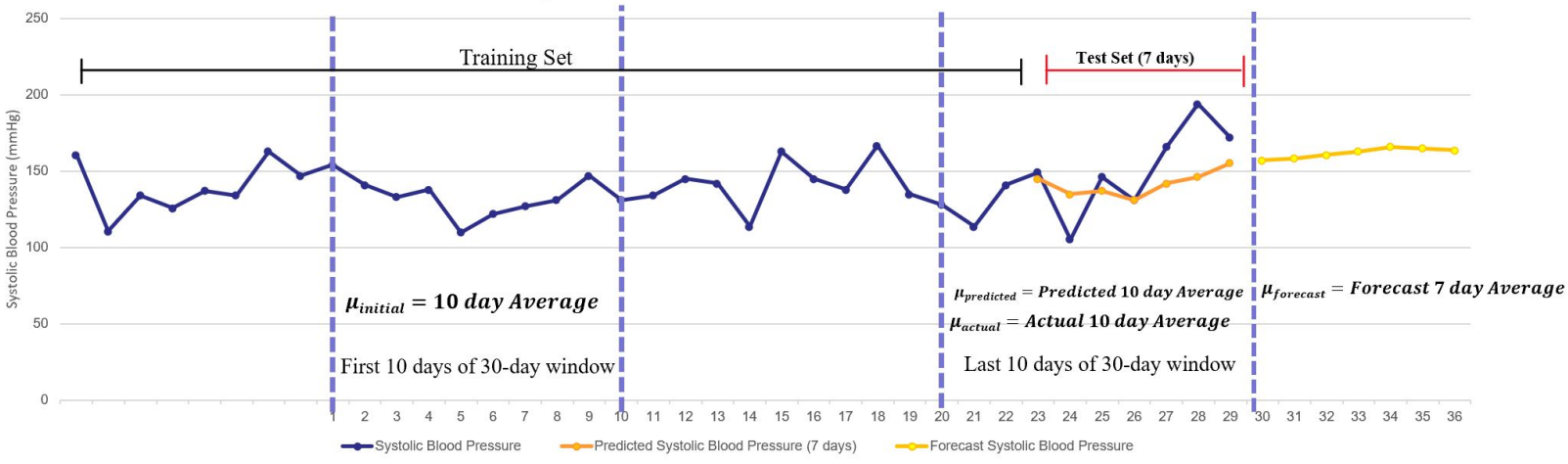
2. A univariate SARIMA model built with StatsModels to account for seasonality.

3. Multivariate VAR model built with StatsModels to account for interaction between vital signs.

Performance Evaluation

Performance of was evaluated by the Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and Root Mean Square Error (RSME) for each patient, by model.

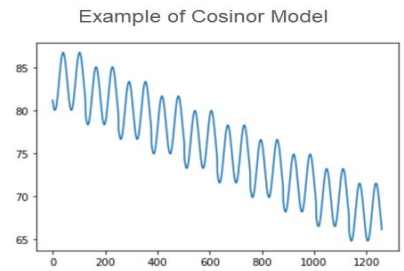
Example of Prediction and Performance Metric

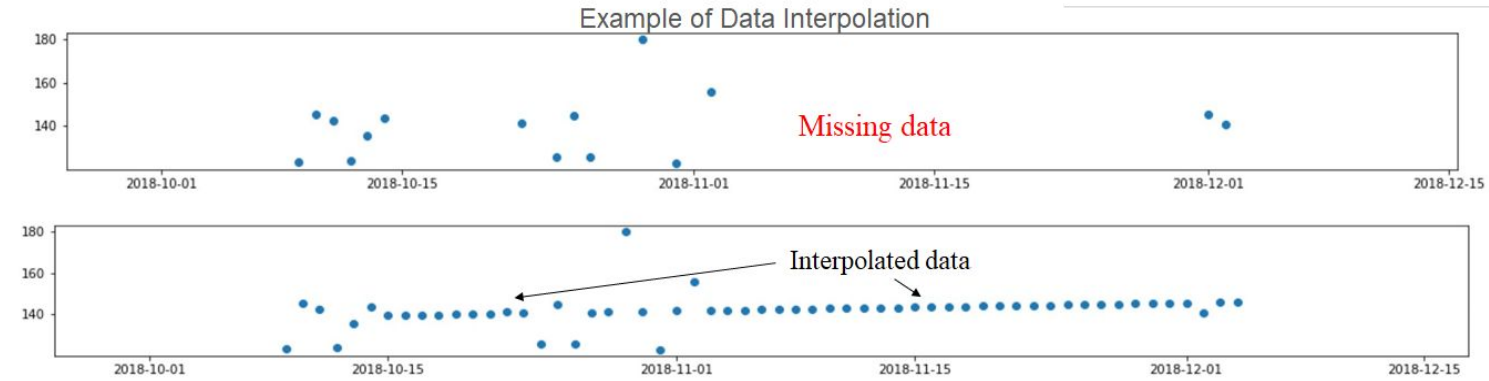


For HR: $Y = M + a\cos\theta + b\sin\theta + c\cos 2\theta + d\sin 2\theta + e\log\Delta$

For SBP: $Y = M + a\cos\theta + b\sin\theta + c\cos 2\theta + d\sin 2\theta + e\cos 3\theta + f\sin 3\theta + g\cos 6\theta + \hbar\sin 6\theta + i\log\Delta$

Where M = MESOR, an estimate of the average value around which the variable oscillates; a,b,c,d,e,f,g = coefficients of fit model; Δ= days elapsed. See figure 1, 2 for sample cosinor model and interpolation





Flags indicate health deterioration, raised for 30-day continuous data with heart rate/bp events that cross risk thresholds or show significant variability.

Flagging algorithm is evaluated by comparing output using predicted and actual values. The metrics used are accuracy and F-1 score for binary flags, MAE for numerical flags.

Results and Conclusions

Table 1 summarizes the performance evaluated using MAPE, MAE and RSME of the three models.

- LSTM model performs well with predicting SBP.
-VAR model performs best when predicting HR, but not as well when predicting SBP because SBP generally did not pass the cointegration test.

The performance of the flag-raising algorithm was also evaluated, as seen on Table 2. VAR model has the highest accuracy and F-1 scores and lowest MAE the three models.

Table1: Model performance metrics.

	LSTM	VAR	SARIMA
HR_mape	0.03	0.04	0.05
HR_mae	4.40	3.22	6.13
HR_rsme	5.47	4.05	7.75
SBP_mape	0.03	0.05	0.05
SBP_mae	2.57	5.96	3.56
SBP_rsme	3.21	7.48	4.58

Table 2: Flag metrics

Flags	LSTM	VAR	SARIMA
acc_rising_HR	0.02	0.99	0.02
f1_rising_HR	0.01	0.50	0.01
acc_lowering_HR	0.98	0.99	0.98
f1_lowering_HR	0.00	0.67	0.00
acc_rising_SBP	0.89	0.91	0.89
f1_rising_SBP	0.02	0.57	0.00
acc_high_BP	0.77	0.93	0.78
f1_high_BP	0.00	0.84	0.00
mae_min_HR	0.27	0.15	0.26
mae_max_HR	6.81	0.10	6.71
mae_min_SBP	6.50	0.06	6.38
mae_max_SBP	0.02	0.04	0.03

Next Steps

Incorporate beta blockers data with half-life of each medicine for a better understanding of deterioration of health in relation to patient medication [2].

Refine data visualization to be operationalized by Sentinel Healthcare platform, including multiple readings per day per patient in a line chart mixed with a scatter showing vital sign means.

References

[1] Bangalore S., et al. "Relation of Beta-Blocker–Induced Heart Rate Lowering and Cardioprotection in Hypertension." *J Am Coll Cardiol*, 2008 Oct, 52 (18) 1482-1489. doi: 10.1016/j.jacc.2008.06.048

[2] Cullington, D., et al. (2012) "Heart rate achieved or beta-blocker dose in patients with chronic heart failure: which is the better target?" *European Journal of Heart Failure*, 2012 14: 737-747. doi:10.1093/eurjhf/hfs060

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